A Deletion Formula for Inverse Kazhdan–Lusztig Polynomials of Matroids

Tom Braden, Luis Ferroni, Jacob P. Matherne, and Nutan Nepal

Department of Mathematics, North Carolina State University

nnepal2@ncsu.edu

Polynomial invariants of matroids

To a loopless matroid M, there are several important associated polynomial invariants:

1. The characteristic polynomial $\chi_{\mathsf{M}}(x)$,
2. The Chow polynomial $\underline{\mathrm{H}}_{\mathsf{M}}(x)$ [7],
3. The Kazhdan–Lusztig polynomial $\mathrm{P}_{\mathsf{M}}(x)$ [5],
4. The inverse Kazhdan–Lusztig polynomial $\mathrm{Q}_{\mathsf{M}}(x)$ [8].

These polynomials have the following algebraic interpretations:

1. $\chi_{\mathsf{M}}(x)$ encodes the Poincaré polynomial of the Orlik–Solomon algebra of M,
2. $\underline{\mathrm{H}}_{\mathsf{M}}(x)$ is the Poincaré polynomial of the Chow ring of M [6, 7],
3. $\mathrm{P}_{\mathsf{M}}(x)$ is the Poincaré polynomial of the local intersection cohomology module of M [3],
4. The coefficient of x^k in $\mathrm{Q}_{\mathsf{M}}(x)$ is the multiplicity of the trivial module in the degree $\mathrm{rk}(\mathsf{M}) - 2k$ piece of the Rouquier complex of M [3].

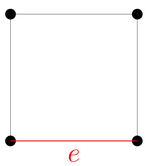
Definition. The inverse Kazhdan–Lusztig polynomial $\mathrm{Q}_{\mathsf{M}}(x)$ of a loopless matroid M is the unique polynomial satisfying

$$\sum_{F \in \mathcal{L}(\mathsf{M})} (-1)^{\mathrm{rk}(F)} \mathrm{Q}_{\mathsf{M}|_F}(x) \cdot \mathrm{P}_{\mathsf{M}/F}(x) = \delta_{\mathsf{M}}$$

which is 1 if M is the empty matroid and 0 otherwise.

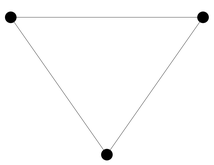
Examples

Let M be the uniform matroid $U_{3,4}$, the graphic matroid on the cycle C_4.



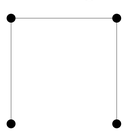
1. $\chi_{\mathsf{M}}(x) = x^3 - 4x^2 + 6x - 3$.
2. $\mathrm{P}_{\mathsf{M}}(x) = 2x + 1$.
3. $\mathrm{Q}_{\mathsf{M}}(x) = 2x + 3$.

After contracting the edge e, we get the matroid M/e on C_3:



1. $\chi_{\mathsf{M}/e}(x) = x^2 - 3x + 2$.
2. $\mathrm{P}_{\mathsf{M}/e}(x) = 1$.
3. $\mathrm{Q}_{\mathsf{M}/e}(x) = 2$.

After deleting the edge e, we get the matroid $\mathsf{M} \setminus e$ on the tree:



1. $\chi_{\mathsf{M}\setminus e}(x) = x^3 - 3x^2 + 3x - 1$.
2. $\mathrm{P}_{\mathsf{M}\setminus e}(x) = 1$.
3. $\mathrm{Q}_{\mathsf{M}\setminus e}(x) = 1$.

Deletion formulas for the invariants

The aforementioned invariants satisfy the following formulas:

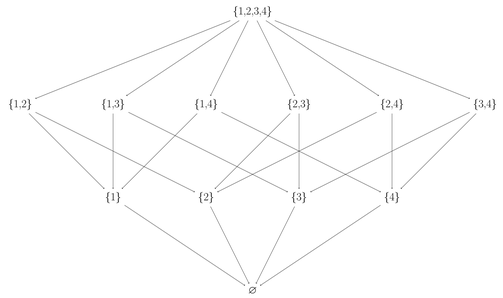
1. $\chi_{\mathsf{M}}(x) = \chi_{\mathsf{M}\setminus e}(x) - \chi_{\mathsf{M}/e}(x)$.
2. $\mathrm{P}_{\mathsf{M}}(x) = \mathrm{P}_{\mathsf{M}\setminus e}(x) - x\mathrm{P}_{\mathsf{M}/e}(x) + \sum_{F \in S} \tau(\mathsf{M}/F \cup e) x^{\mathrm{crk}(F)/2} \cdot \mathrm{P}_{\mathsf{M}|_F}(x)$

where $S := \{F \in \mathcal{L}(\mathsf{M}) \mid e \notin F \text{ and } F \cup e \in \mathcal{L}(\mathsf{M})\}$ and $\tau(\mathsf{M})$ is the coefficient of $x^{\lfloor \mathrm{rk}(\mathsf{M})/2 \rfloor}$ in the $\mathrm{P}_{\mathsf{M}}(x)$ [4].

Main theorem [2]. The inverse Kazhdan–Lusztig polynomial satisfies the following deletion formula:

$$\mathrm{Q}_{\mathsf{M}}(x) = \mathrm{Q}_{\mathsf{M}\setminus e}(x) + (1+x)\mathrm{Q}_{\mathsf{M}/e}(x) - \sum_{F \in T} \tau(\mathsf{M}|_F/e) x^{\mathrm{rk}(F)/2} \cdot \mathrm{Q}_{\mathsf{M}/F}(x)$$

where $T := \{F \in \mathcal{L}(\mathsf{M}) \mid e \in F \text{ and } F \setminus e \notin \mathcal{L}(\mathsf{M})\}$.



From the lattice of flats of $U_{3,4}$ above, we can see that the set T consists of just the top flat for which $\tau(\mathsf{M}|_F/e) = 0$ and the deletion formula reduces to

$$\mathrm{Q}_{\mathsf{M}}(x) = 1 + (1+x) \cdot 2 - 0 = 2x + 3.$$

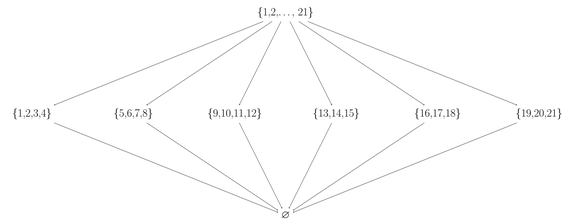
Application: Counterexample to a log-concavity conjecture

A polynomial $\sum_{i=0}^{d} a_i x^i \in \mathbb{R}_{\geq 0}[x]$ is log-concave if $a_k^2 \geq a_{k-1}a_{k+1}$ for all $0 < k < d$. The log-concavity of the polynomial invariants of matroids is a topic of ongoing research.

1. The log-concavity of $\chi_{\mathsf{M}}(x)$ (Heron–Rota–Welsh conjecture) was proved by Adiprasito, Huh, and Katz in [1].
2. **Conjecture:** The polynomials $\mathrm{P}_{\mathsf{M}}(x)$ and $\mathrm{Q}_{\mathsf{M}}(x)$ are log-concave [9, 8].

In their proof of the log-concavity of the inverse Kazhdan–Lusztig polynomial of paving matroids, Xie and Zhang conjectured that the Hadamard product of the inverse Kazhdan–Lusztig polynomial of degree d with the polynomial $(1+x)^d$ is real-rooted. This conjecture, if true, would imply the log-concavity of the inverse Kazhdan–Lusztig polynomial of all matroids.

We use the deletion formula to provide a counterexample to this conjecture. Let M be the rank 2 matroid on the ground set $\{1, 2, \ldots, 21\}$.



Using the deletion formula, we can compute the inverse Kazhdan–Lusztig polynomial of the dual matroid M^*. We find the Hadamard product $\mathrm{Q}_{\mathsf{M}^*}(x) \star (1+x)^{\deg(\mathrm{Q}_{\mathsf{M}^*}(x))}$ to be

$$71162x^9 + 2093958x^8 + 12614544x^7 + 27186264x^6 + 27111294x^5 + 13439034x^4 + 3294228x^3 + 371628x^2 + 16110x + 163$$

which we can see, by using SageMath, is not real-rooted. Note that this does not contradict the conjecture regarding the log-concavity of the polynomials $\mathrm{P}_{\mathsf{M}}(x)$ and $\mathrm{Q}_{\mathsf{M}}(x)$.

References

[1] K. Adiprasito, J. Huh, and E. Katz. "Hodge theory for combinatorial geometries". In: *Ann. of Math. (2)* 188.2 (2018), pp. 381–452.

[2] T. Braden, L. Ferroni, J. P. Matherne, and N. Nepal. *Inverse Kazhdan-Lusztig polynomials of matroids under deletion*. 2025. arXiv: 2510.01086.

[3] T. Braden, J. Huh, J. P. Matherne, N. Proudfoot, and B. Wang. *Singular Hodge theory for combinatorial geometries*. Oct. 2022. arXiv: 2010.06088.

[4] T. Braden and A. Vysogorets. "Kazhdan-Lusztig polynomials of matroids under deletion". In: *Electron. J. Combin.* 27.1 (2020), Paper No. 1.17, 17.

[5] B. Elias, N. Proudfoot, and M. Wakefield. "The Kazhdan-Lusztig polynomial of a matroid". In: *Adv. Math.* 299 (2016), pp. 36–70.

[6] E. M. Feichtner and S. Yuzvinsky. "Chow rings of toric varieties defined by atomic lattices". In: *Invent. Math.* 155.3 (2004), pp. 515–536.

[7] L. Ferroni, J. P. Matherne, M. Stevens, and L. Vecchi. "Hilbert-Poincaré series of matroid Chow rings and intersection cohomology". In: *Adv. Math.* 449 (2024), Paper No. 109733.

[8] A. L. L. Gao and M. H. Y. Xie. "The inverse Kazhdan-Lusztig polynomial of a matroid". In: *J. Combin. Theory Ser. B* 151 (2021), pp. 375–392.

[9] K. Gedeon, N. Proudfoot, and B. Young. "Kazhdan-Lusztig polynomials of matroids: a survey of results and conjectures". In: *Sém. Lothar. Combin.* 78B (2017), Art. 80, 12.